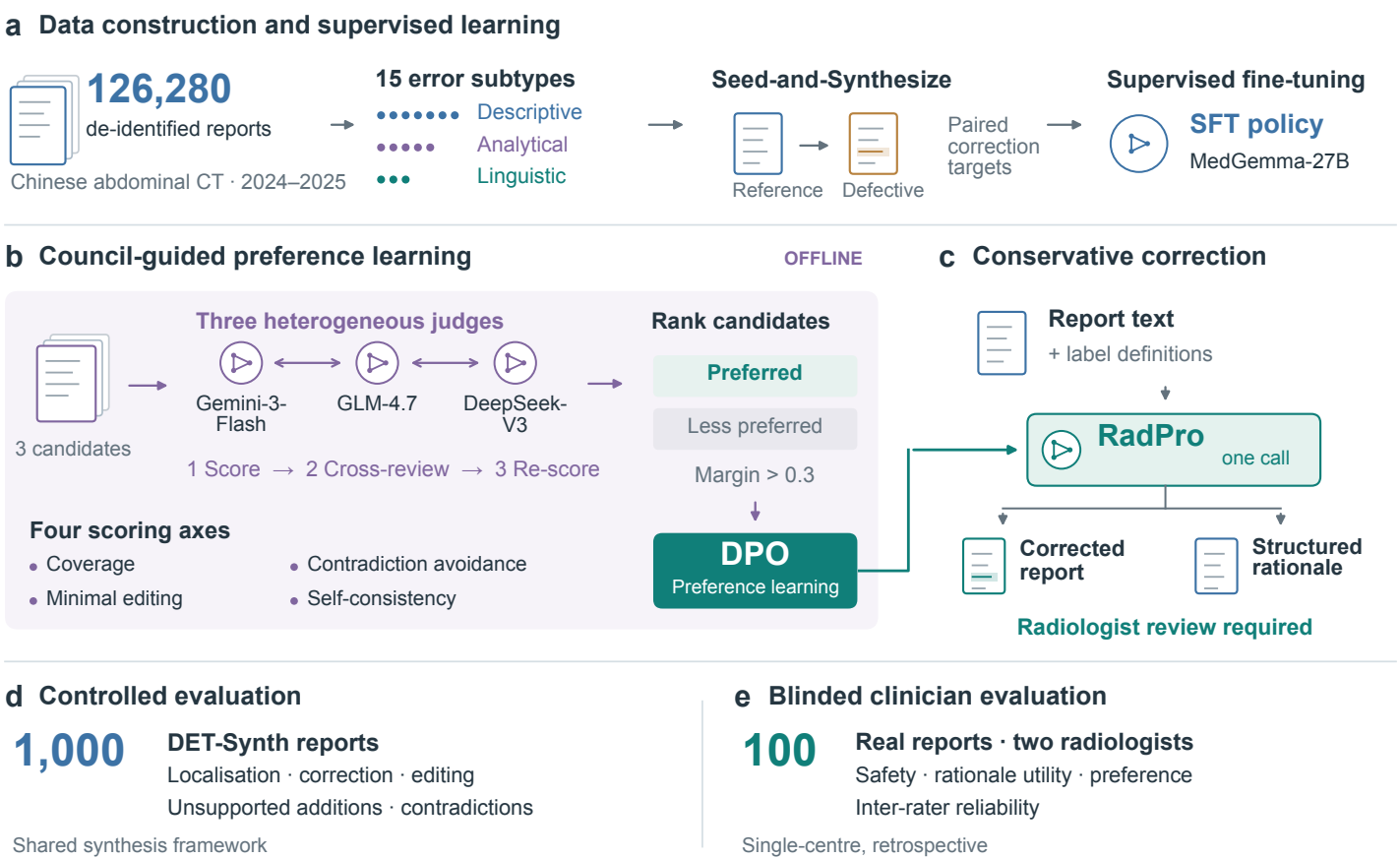
a Data construction and supervised learning
126,280
de-identified reports
Chinese abdominal CT · 2024–2025
15 error subtypes
Descriptive
Analytical
Linguistic
Seed-and-Synthesize
Reference
Defective
Paired correction targets
Supervised fine-tuning
SFT policy
MedGemma-27B
b Council-guided preference learning
OFFLINE
3 candidates
Three heterogeneous judges
Gemini-3-Flash
GLM-4.7
DeepSeek-V3
1 Score → 2 Cross-review → 3 Re-score
Four scoring axes
Coverage
Minimal editing
Contradiction avoidance
Self-consistency
Rank candidates
Preferred
Less preferred
Margin > 0.3
DPO
Preference learning
c Conservative correction
Report text
+ label definitions
RadPro
one call
Corrected report
Structured rationale
Radiologist review required
d Controlled evaluation
1,000
DET-Synth reports
Localisation · correction · editing
Unsupported additions · contradictions
Shared synthesis framework
e Blinded clinician evaluation
100
Real reports · two radiologists
Safety · rationale utility · preference
Inter-rater reliability
Single-centre, retrospective